

CSE241 – Security of Information Systems

Spring 2026

Lecture (3)

Cryptographic Tools

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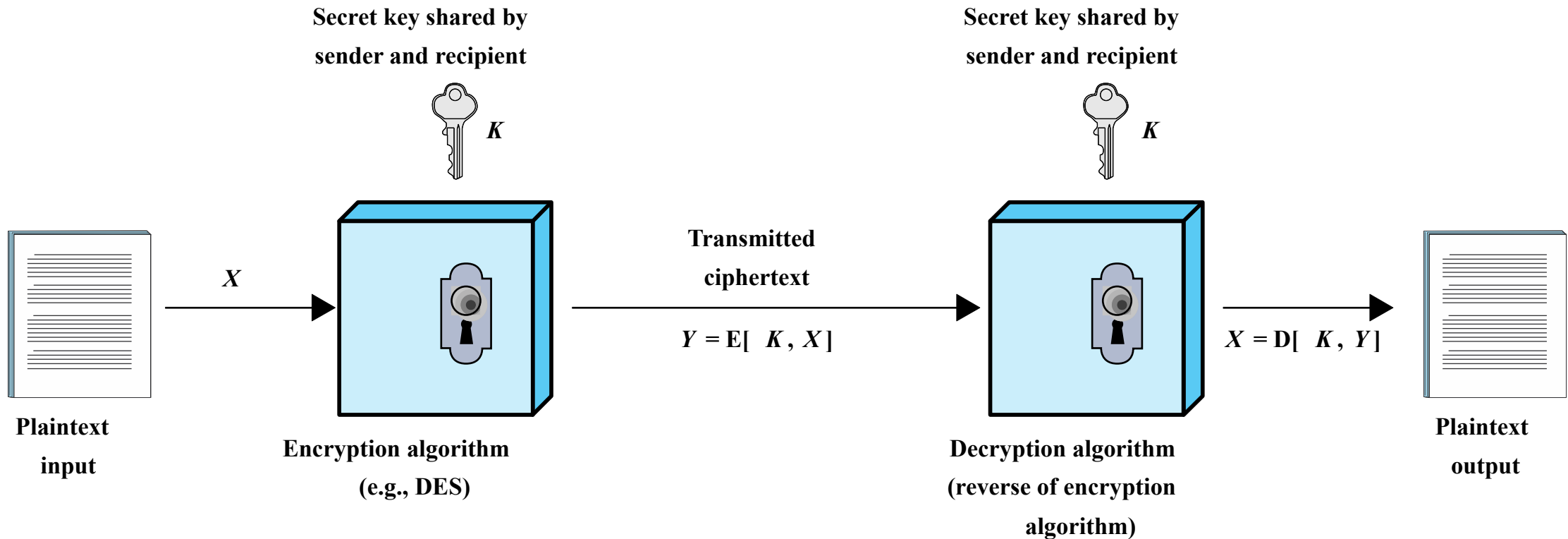
Objectives

- Explain the basic operation of symmetric encryption algorithms.
- Compare and contrast block and stream encryption algorithms.
- Discuss the use of secure hash functions for message authentication.
- List other applications of secure hash functions.
- Explain the basic operation of asymmetric block encryption algorithms.
- Present an overview of the digital signature mechanism and explain the concept of digital envelopes.
- Explain the significance of random and pseudorandom numbers in cryptography.

Symmetric Encryption

- The universal technique for providing confidentiality for transmitted or stored data.
- Also referred to as conventional encryption or single-key encryption.
- Two requirements for secure use:
 - Need a **strong** encryption algorithm = *more resistant to cryptanalysis*
 - Sender and receiver must **exchange** the secret key in a secure fashion
 - Sender and receiver must keep the key secure, **How** ?

Simplified Model of Symmetric Encryption



Attacking Symmetric Encryption

Cryptanalytic Attacks

- Rely on:
 - Nature of the algorithm
 - Some knowledge of the general characteristics of the plaintext
 - Some sample plaintext-ciphertext pairs
- Exploits the characteristics of the algorithm to attempt to deduce a specific plaintext or the key being used
 - If successful all **future** and **past** messages encrypted with that key are compromised

Brute-Force Attacks

- Try all possible keys on some ciphertext until an intelligible translation into plaintext is obtained
 - On average half of all possible keys must be tried to achieve success

Comparison of Three Popular Symmetric Encryption Algorithms

	DES	Triple DES	AES
Plaintext block size (bits)	64	64	128
Ciphertext block size (bits)	64	64	128
Key size (bits)	56	112 or 168	128, 192, or 256

DES = Data Encryption Standard

AES = Advanced Encryption Standard

Data Encryption Standard (DES) (1975 – 2004)



Until recently was the most widely used encryption scheme

- FIPS PUB 46 (1977)
- Referred to as the Data Encryption Algorithm (DEA)
- Uses **64-bit plaintext block** and **56-bit key** to produce a **64-bit ciphertext block**

Strength concerns:

- Concerns about the algorithm itself
 - DES is the most studied encryption algorithm in existence
- Concerns about the use of a 56-bit key
 - The speed of commercial off-the-shelf processors makes this key length inadequate

Average Time Required for Exhaustive Key Search

Key Size (bits)	Cipher	Number of Alternative Keys	Time Required at 10^9 decryptions/ μs	Time Required at 10^{13} decryptions/ μs
56	DES	$2^{56} \approx 7.2 \times 10^{16}$	$2^{55} \mu s = 1.125$ years	1 hour
128	AES	$2^{128} \approx 3.4 \times 10^{38}$	$2^{127} \mu s = 5.3 \times 10^{21}$ years	5.3×10^{17} years
168	Triple DES	$2^{168} \approx 3.7 \times 10^{50}$	$2^{167} \mu s = 5.8 \times 10^{33}$ years	5.8×10^{29} years
192	AES	$2^{192} \approx 6.3 \times 10^{57}$	$2^{191} \mu s = 9.8 \times 10^{40}$ years	9.8×10^{36} years
256	AES	$2^{256} \approx 1.2 \times 10^{77}$	$2^{255} \mu s = 1.8 \times 10^{60}$ years	1.8×10^{56} years

Triple DES (3DES) (1981-2024)

- Repeats basic DES algorithm three times using either two or three unique keys.
- First standardized for use in financial applications in ANSI standard X9.17 in 1985.
- Attractions:
 - **168-bit key** length overcomes the vulnerability to brute-force attack of DES
 - Underlying encryption algorithm is the same as in DES
- Drawbacks:
 - Slow, inefficient code
 - Uses a 64-bit block size

Advanced Encryption Standard (AES)

Needed a replacement
for 3DES

3DES was not
reasonable for
long term use

NIST called for
proposals for a new
AES in 1997

Should have a security
strength equal to or better
than 3DES

Significantly improved
efficiency

Symmetric block cipher

128-bit data and
128/192/256-bit keys

Selected Rijndael in
November 2001

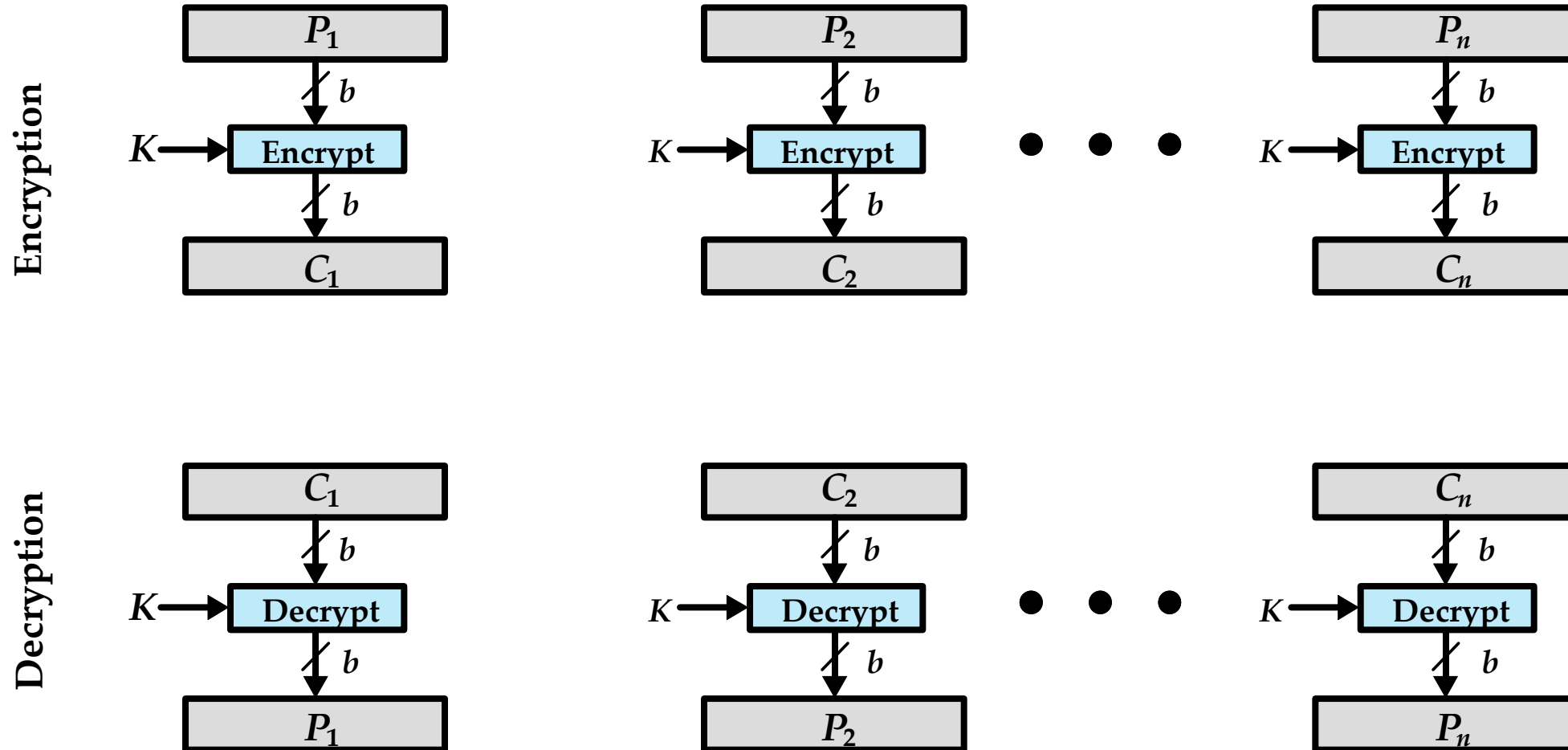
Published as
FIPS 197

Practical Security Issues

- Typically, symmetric encryption is applied to a unit of data larger than a single 64-bit or 128-bit block
- Electronic codebook (ECB) mode is the simplest approach to multiple-block encryption
 - Each block of plaintext is encrypted using the same key
 - Cryptanalysts may be able to exploit regularities in the plaintext
- Modes of operation such as (CBC, CFB, OFB, CTR)
 - Alternative techniques developed to increase the security of symmetric block encryption for large sequences
 - Overcomes the weaknesses of ECB

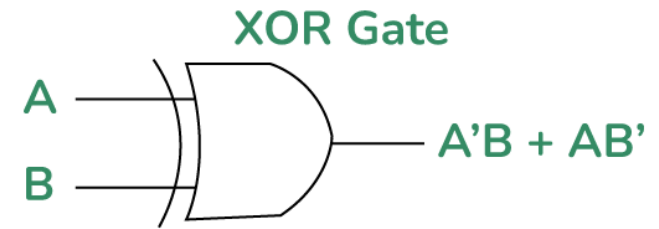
Types of Symmetric Encryption

Block cipher encryption ECB mode



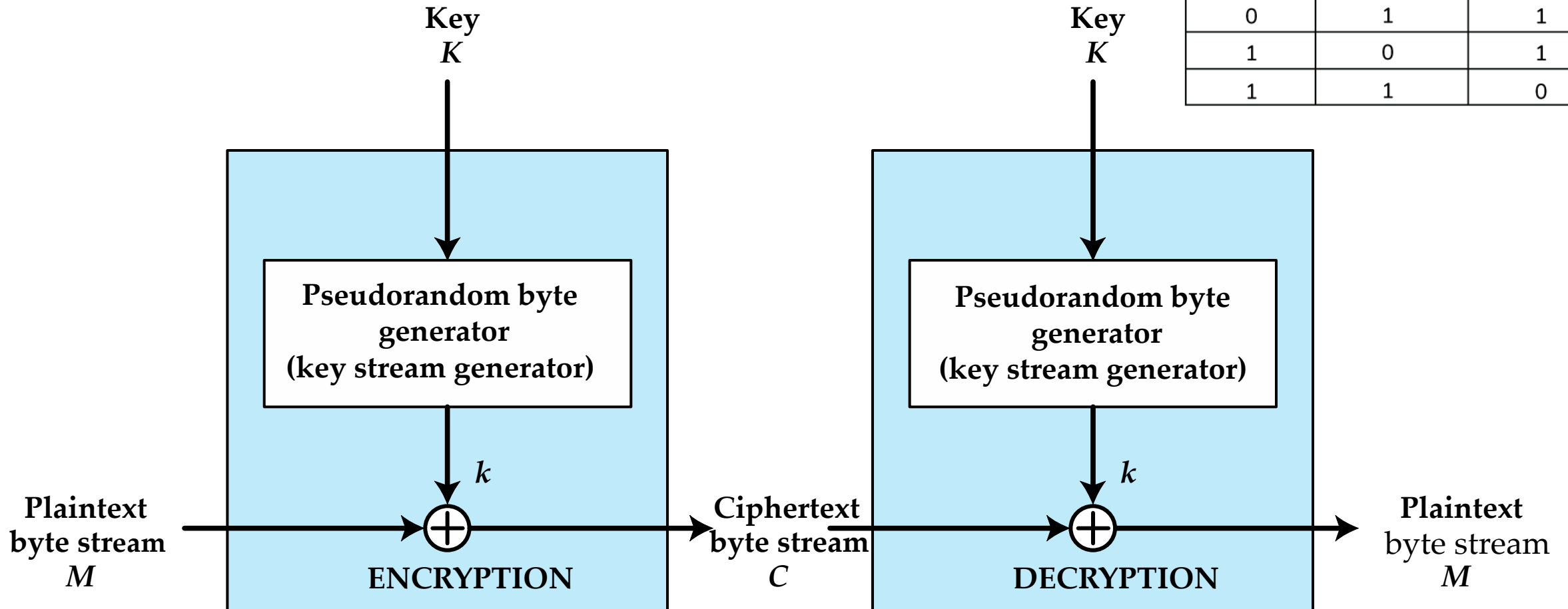
Types of Symmetric Encryption

Stream encryption



Truth Table

A (Input 1)	B (Input 2)	$X = A'B + AB'$
0	0	0
0	1	1
1	0	1
1	1	0



Block & Stream Ciphers

Block Cipher

- Processes the input one block of elements at a time
- Produces an output block for each input block
- Can reuse keys
- More common

Stream Cipher

- Processes the input elements continuously
- Produces output one element at a time
- Primary advantage is that they are almost always faster and use far less code
- Encrypts plaintext one byte at a time
- Pseudorandom stream is one that is unpredictable without knowledge of the input key

Message Authentication



Protects against active attacks	
Verifies received message is authentic	• Contents have not been altered • From authentic source • Timely and in correct sequence
Can use conventional encryption	• Only sender and receiver share a key

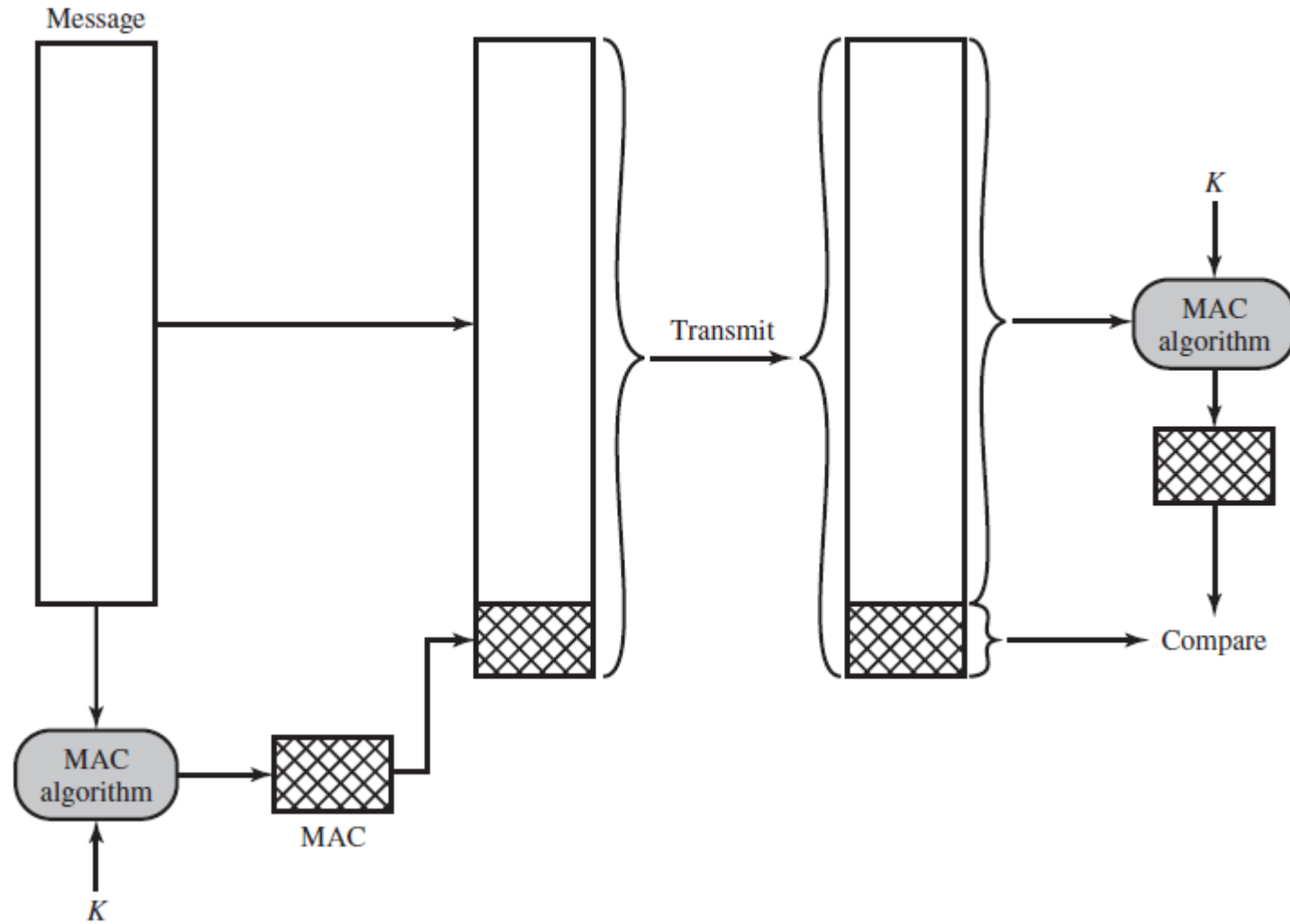
Message Authentication Without Confidentiality

- Message encryption by itself does not provide a secure form of authentication.
- It is possible to combine authentication and confidentiality in a single algorithm by encrypting a message plus its authentication tag.
- Typically, message authentication is provided as a separate function from message encryption.

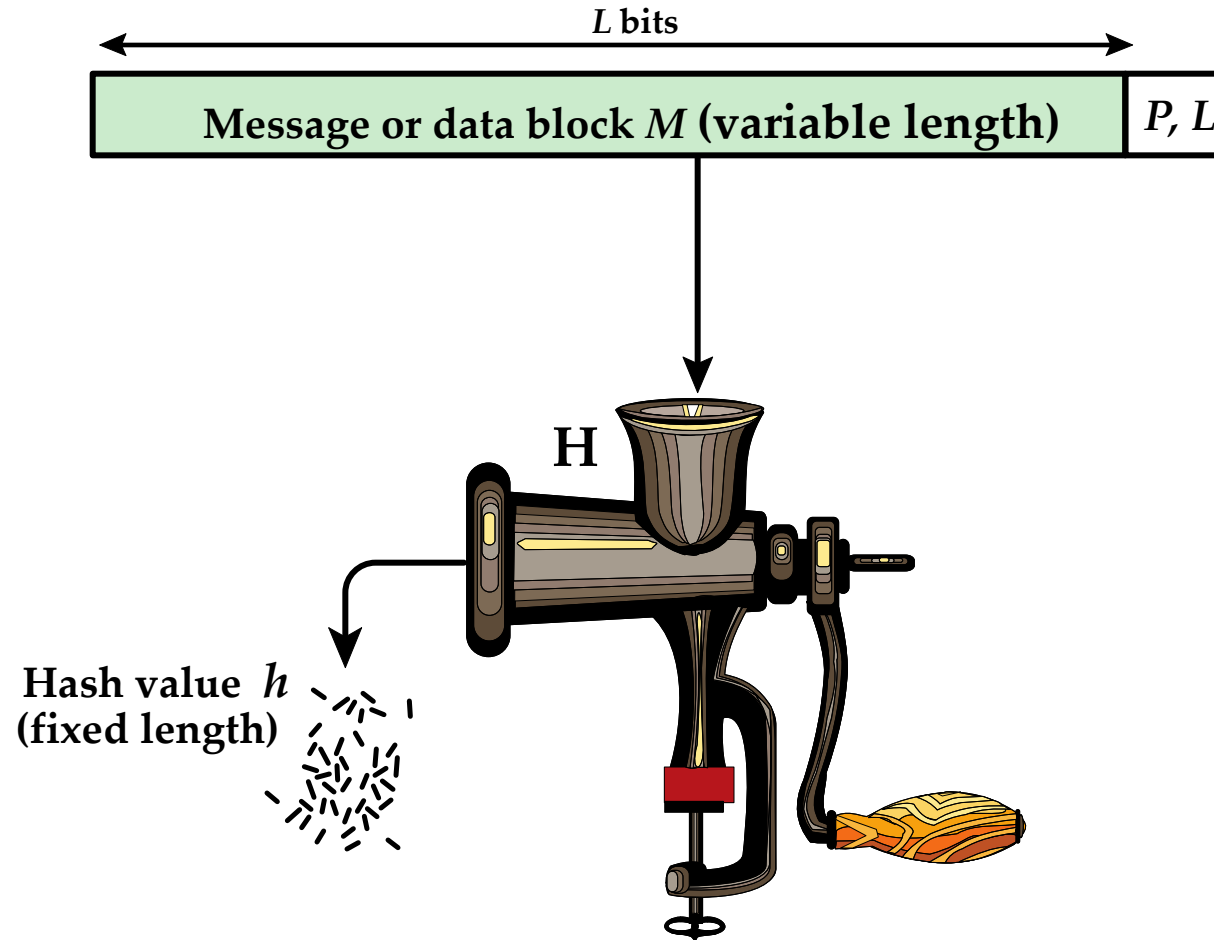
Message Authentication Without Confidentiality (Contd.)

- Situations in which message authentication without confidentiality may be preferable include:
 - There are a number of applications in which the same message is broadcast to a number of destinations
 - An exchange in which one side has a heavy load and cannot afford the time to decrypt all incoming messages
 - Authentication of a computer program in plaintext is an attractive service.

Message Authentication Using a Message Authentication Code (MAC)



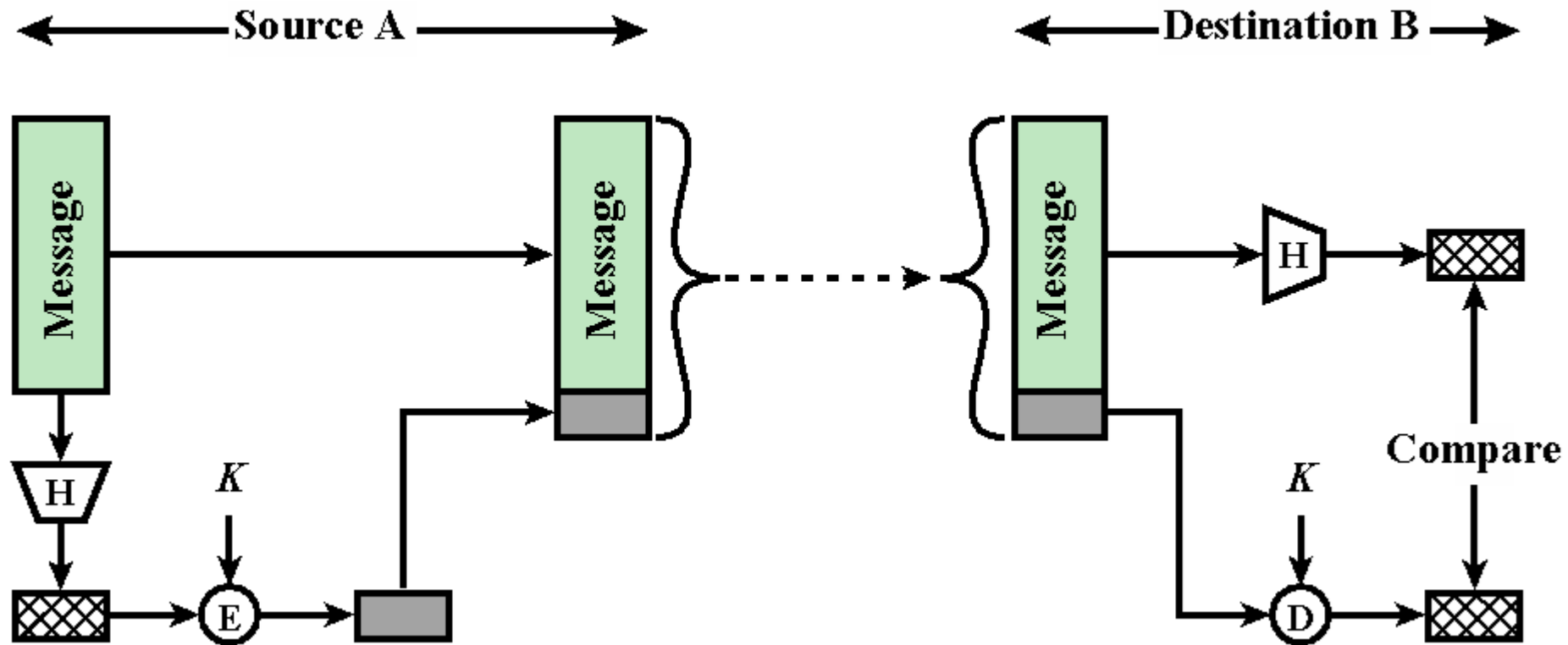
Cryptographic Hash Function; $h = H(M)$



P, L = padding plus length field

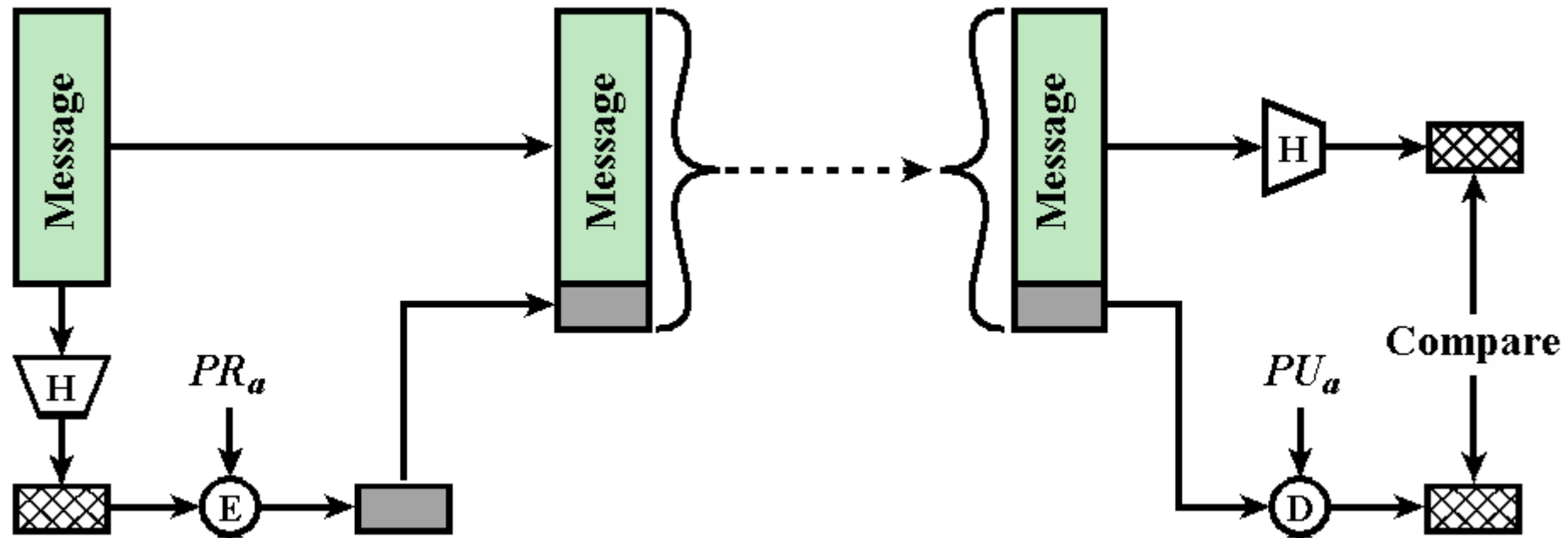
Message Authentication Using a One-Way Hash Function

(a) Using symmetric encryption



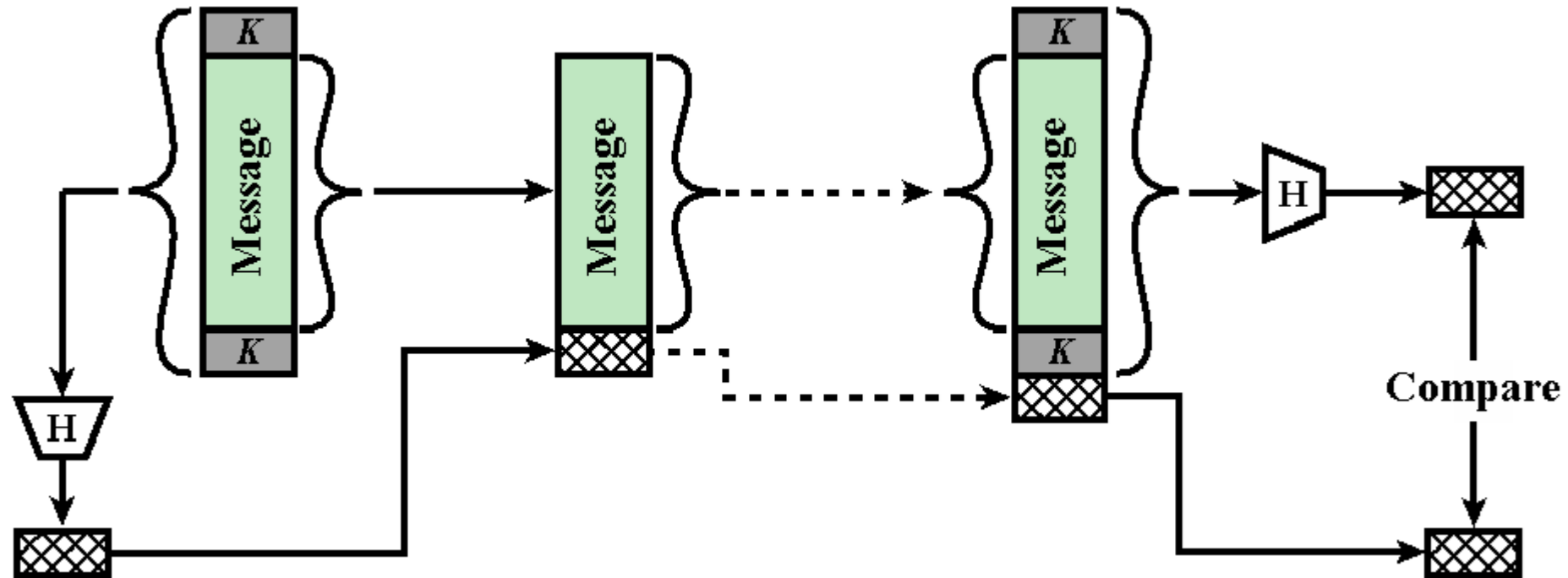
Message Authentication Using a One-Way Hash Function

(b) Using public-key encryption



Message Authentication Using a One-Way Hash Function

(c) Using secret value



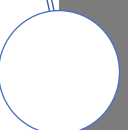
Properties of a Hash function in Message Authentication



Can be applied to a block of data of any size



Produces a fixed-length output



$H(x)$ is relatively easy to compute for any given x



One-way - Computationally infeasible to find x such that $H(x) = h$



Weak Collision Resistance - Computationally infeasible to find $y \neq x$ such that $H(y) = H(x)$



Strong Collision Resistance - Computationally infeasible to find any pair (x, y) such that $H(x) = H(y)$

Security of Hash Functions

There are two approaches to attacking a secure hash function:

Cryptanalysis

- Exploit logical weaknesses in the algorithm

Brute-force attack

- Strength of hash function depends solely on the length of the hash code produced by the algorithm

SHA most widely used hash algorithm

SHA-1 (160 bit)

SHA-2 (256, 384, and 512 bits)

SHA-3 (Keccak)

Additional secure hash function applications:

Passwords

- Hash of a password is stored by an operating system

Intrusion detection

- Store $H(F)$ for each file on a system and secure the hash values

Public-Key Encryption Structure

Publicly
proposed by
Diffie and
Hellman in 1976

Based on
mathematical
functions

Asymmetric

- Uses two separate keys
- Public key and private key
- Public key is made public for others to use

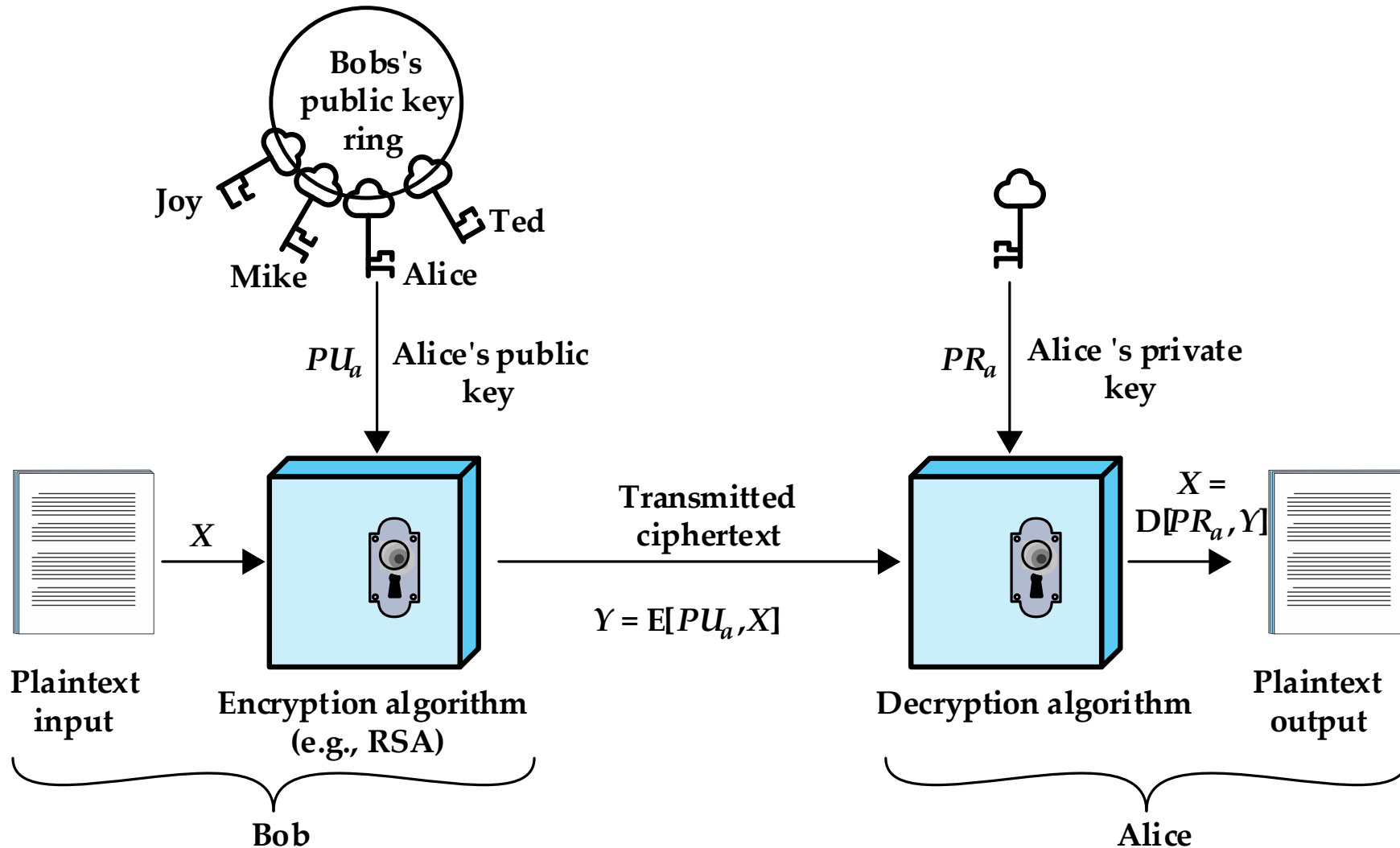
Some form of
protocol is
needed for
distribution

A public-key encryption scheme ingredients

- Plaintext
 - Readable message or data that is fed into the algorithm as input
- Encryption algorithm
 - Performs transformations on the plaintext
- Public and private key
 - Pair of keys, one for encryption, one for decryption
- Ciphertext
 - Scrambled message produced as output
- Decryption key
 - Produces the original plaintext

Public-Key Cryptography

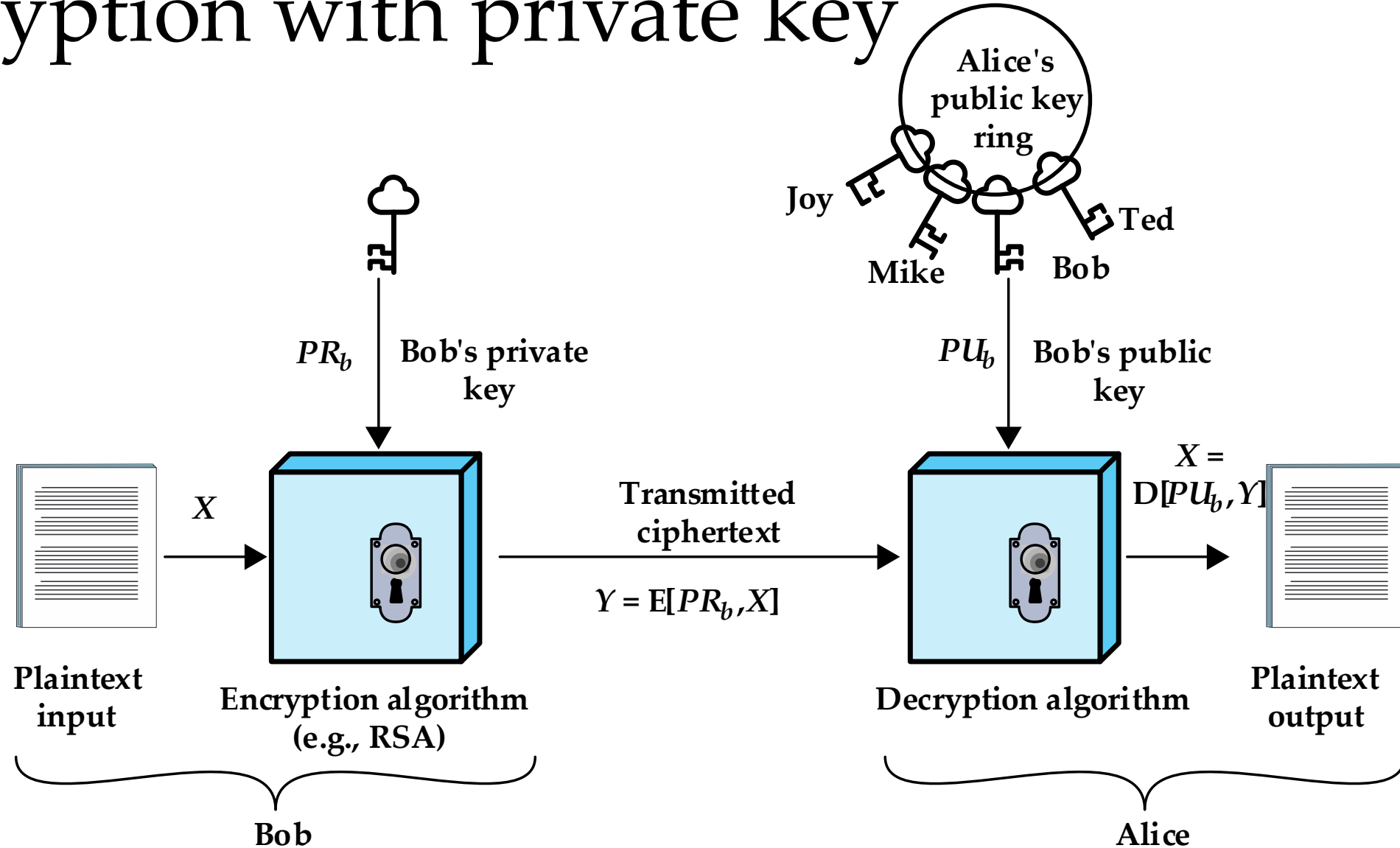
Encryption with public key



Public-Key Cryptography

Encryption with private key

- User encrypts data using his or her own private key
- Anyone who knows the corresponding public key will be able to decrypt the message



Asymmetric Encryption Algorithms

RSA (Rivest, Shamir, Adleman)

Developed in 1976

Most widely accepted and implemented approach to public-key encryption

Diffie-Hellman (DH) key exchange algorithm

Enables two users to securely reach agreement about a shared secret that can be used as a secret key for subsequent symmetric encryption of messages

Limited to the exchange of the keys

Digital Signature Standard (DSS)

Makes use of SHA-1

Provides only a digital signature function with SHA-1

Cannot be used for encryption or key exchange

Elliptic curve cryptography (ECC)

Security like RSA, but with much smaller keys

Reduced processing overhead

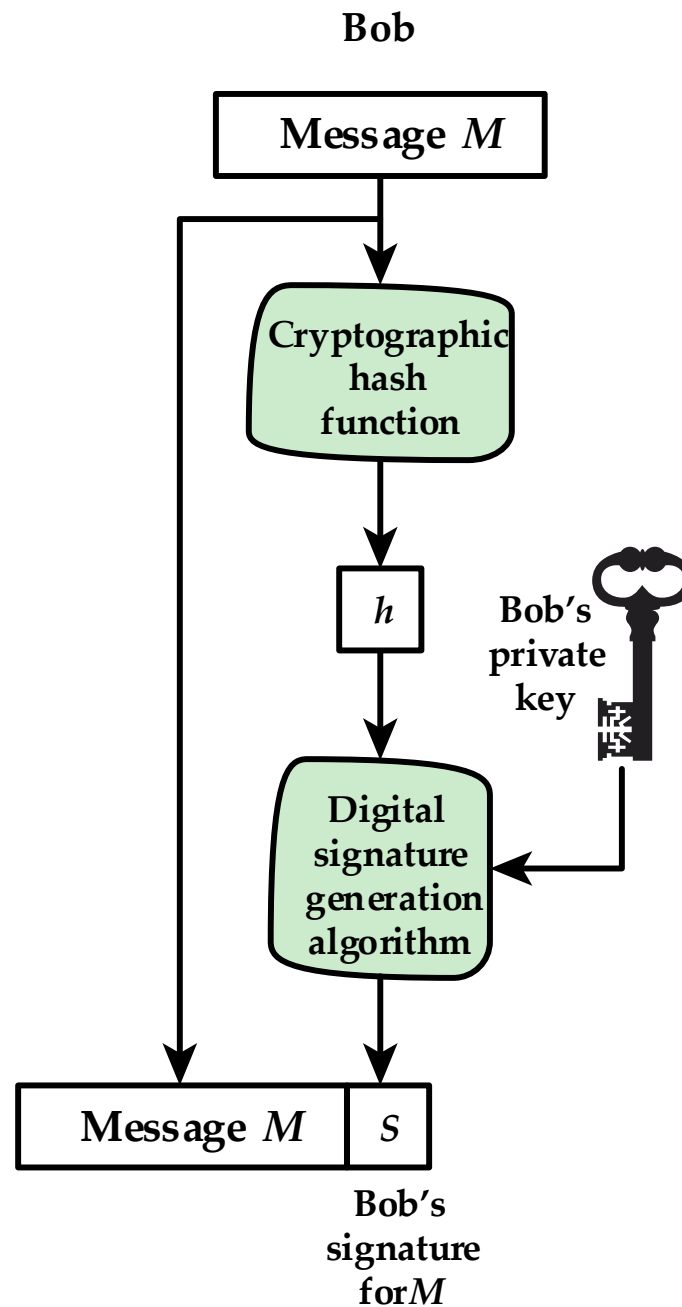
Applications for Public-Key Cryptosystems

Algorithm	Digital Signature	Symmetric Key Distribution	Encryption of Secret Keys
RSA	Yes	Yes	Yes
Diffie-Hellman	No	Yes	No
DSS	Yes	No	No
Elliptic Curve	Yes	Yes	Yes

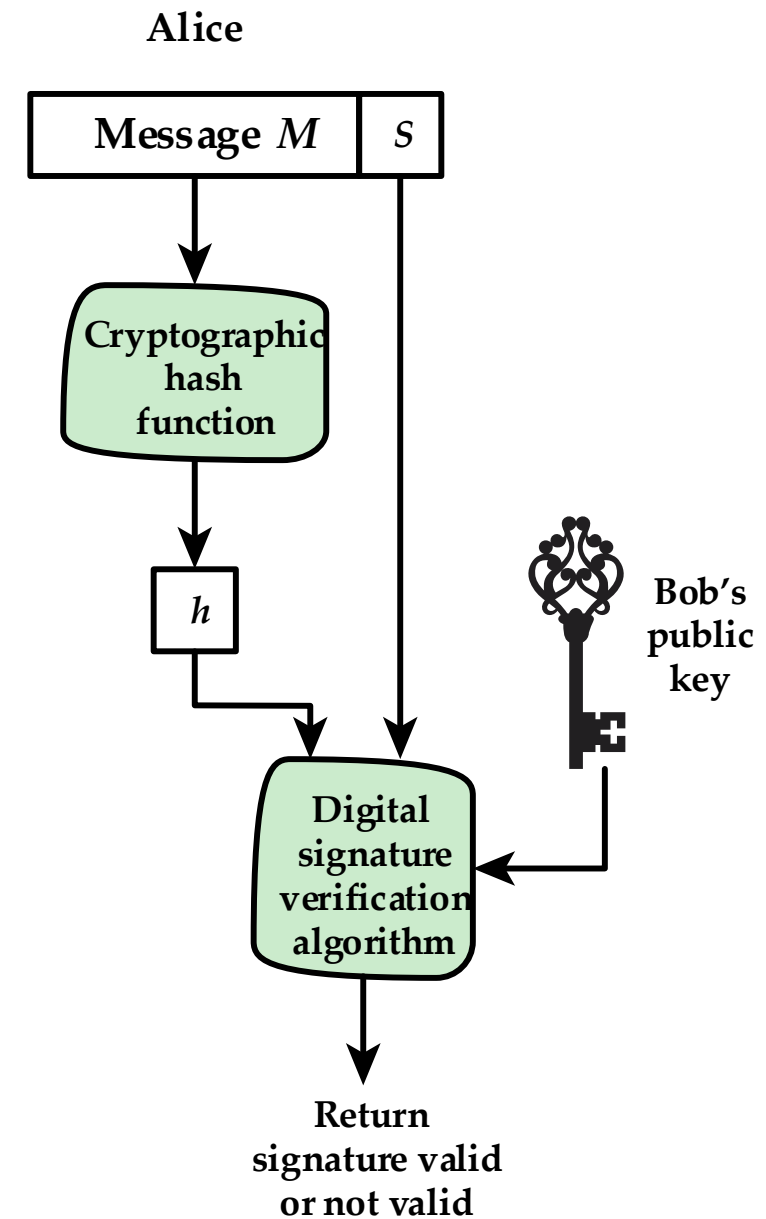
Digital Signatures

- NIST FIPS PUB 186-4 defines a digital signature as:
 - "The result of a **cryptographic transformation of data** that, when properly implemented, provides a mechanism for verifying **origin authentication, data integrity** and signatory **non-repudiation**."
- Thus, a digital signature is a data-dependent bit pattern, generated by an agent as a function of a data block.
- FIPS 186-4 specifies the use of one of three digital signature algorithms:
 - Digital Signature Algorithm (DSA)
 - RSA Digital Signature Algorithm
 - Elliptic Curve Digital Signature Algorithm (ECDSA)

Simplified Depiction of Essential Elements of Digital Signature Process



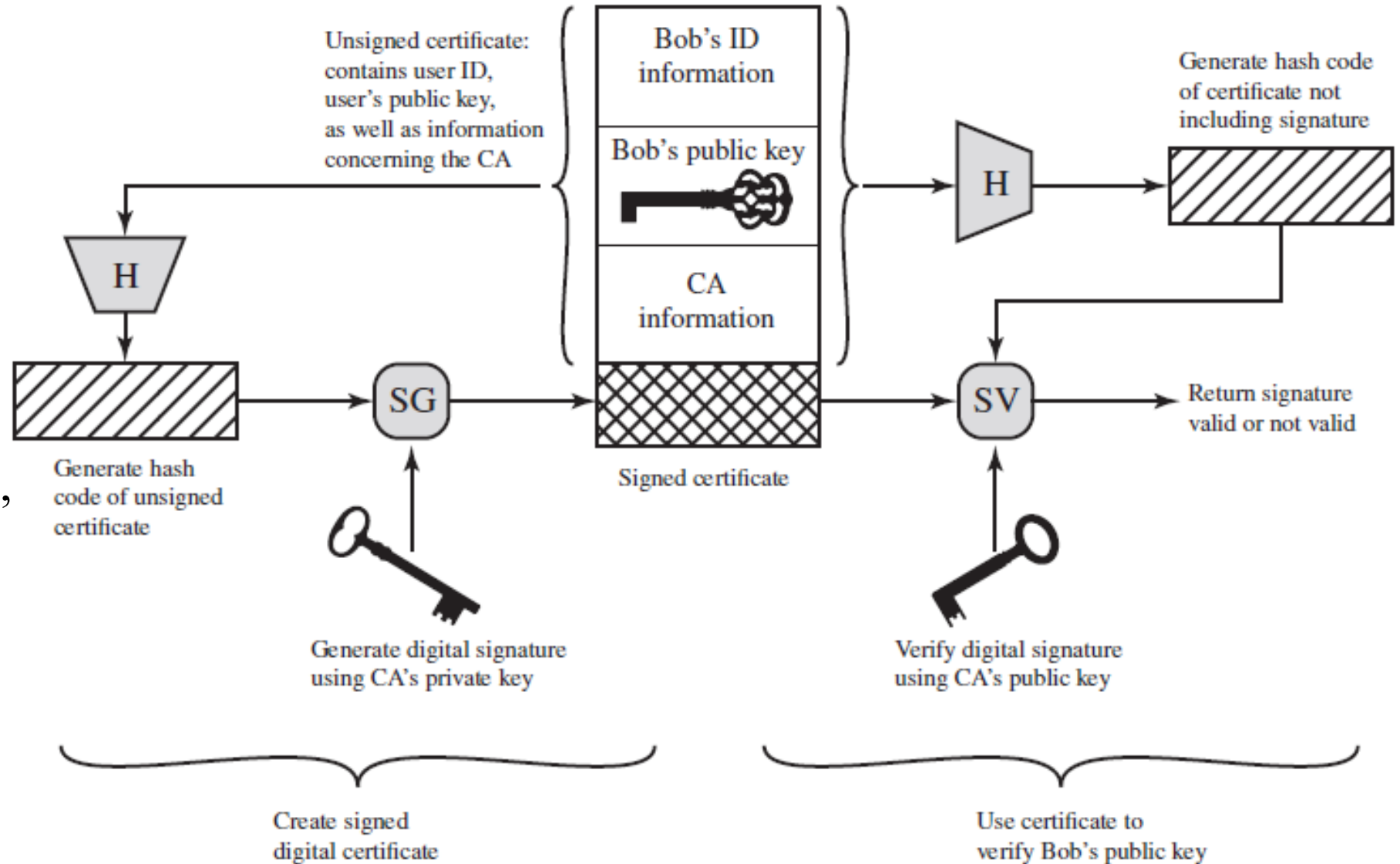
(a) Bob signs a message



(b) Alice verifies the signature

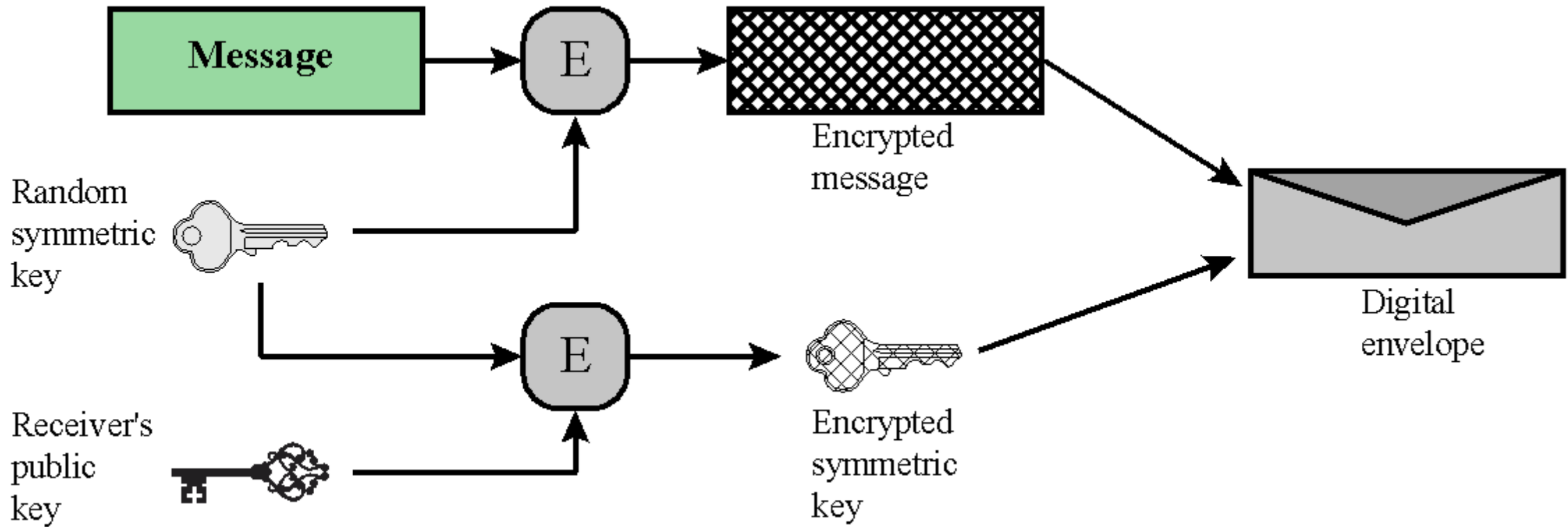
Public-Key Certificate

- X.509 standard.
- Applications:
 - IP Security (IPsec),
 - Transport Layer Security (TLS),
 - Secure Shell (SSH),
 - Secure Multipurpose Internet Mail Extension (S/MIME).



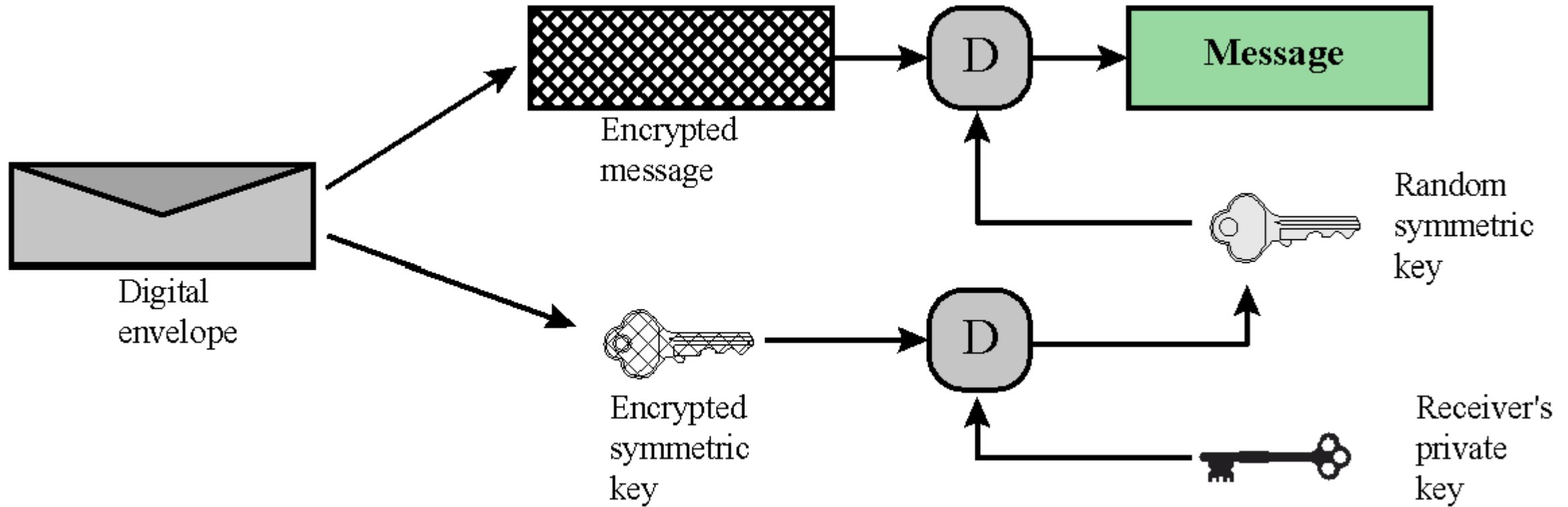
Digital Envelopes

Creation of a digital envelope



Digital Envelopes

Opening a digital envelope



Random Numbers

- Uses include generation of:
 - Keys for public-key algorithms
 - Stream key for symmetric stream cipher
 - Symmetric key for use as a temporary session key or in creating a digital envelope
 - Handshaking to prevent replay attacks
 - Session key

Random Number Requirements

Randomness

- Uniform distribution
 - Frequency of occurrence of each of the numbers should be approximately the same
- Independence
 - No one value in the sequence can be inferred from the others

Unpredictability

- Each number is statistically independent of other numbers in the sequence
- Opponent should not be able to predict future elements of the sequence on the basis of earlier elements

Random versus Pseudorandom

Cryptographic applications typically make use of algorithmic techniques for random number generation

- Algorithms are deterministic and therefore produce sequences of numbers that are not statistically random



Pseudorandom numbers are:

- Sequences produced that satisfy statistical randomness tests
- Likely to be predictable



True random number generator (TRNG):

- Uses a nondeterministic source to produce randomness
- Most operate by measuring unpredictable natural processes
 - e.g. radiation, gas discharge, leaky capacitors
- Increasingly provided on modern processors

Practical Application: Encryption of Stored Data

- Common to encrypt transmitted data
- Much less common for stored data
 - There is often little protection beyond domain authentication and operating system access controls
 - Data are archived for indefinite periods
 - Even though erased, until disk sectors are reused data are recoverable
- Approaches to encrypt stored data:
 - Use a commercially available encryption package
 - Back-end appliance
 - Library based tape encryption
 - Background laptop/PC data encryption

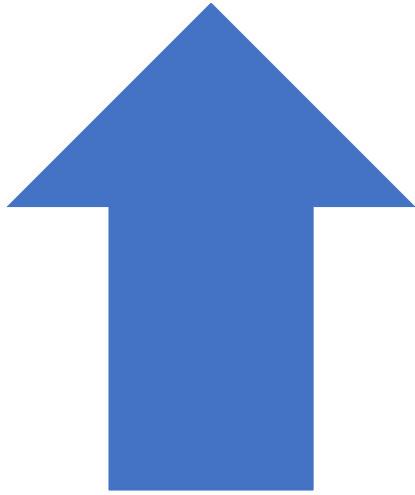
Ethical Aspects in Cryptography

Ethical Aspects in Cryptography

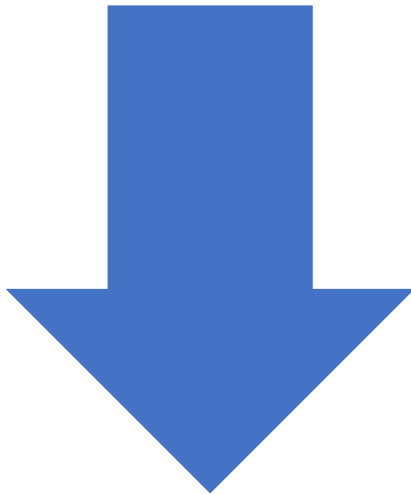
- The use of cryptography raises a wide range of ethical concerns, including:
 - The right to privacy
 - The right to security
 - The right to access
 - The right to equality
 - Potential for abuse
- The ethical concerns of cryptography revolves around:
 - Keys preservation
 - Intellectual property
 - Copyright

Ethical Aspects in Cryptography

The Right to Privacy



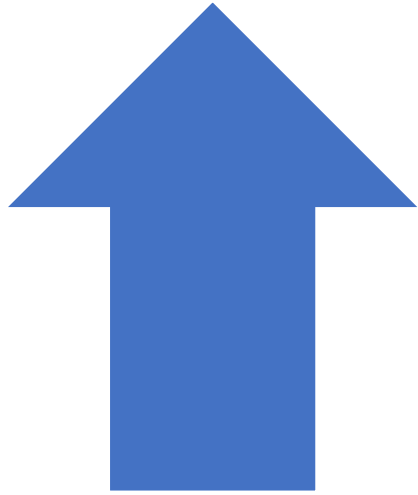
Cryptography is often used to protect the privacy of individuals' data e.g. Personal Identifiable Information (PII) or financial records.



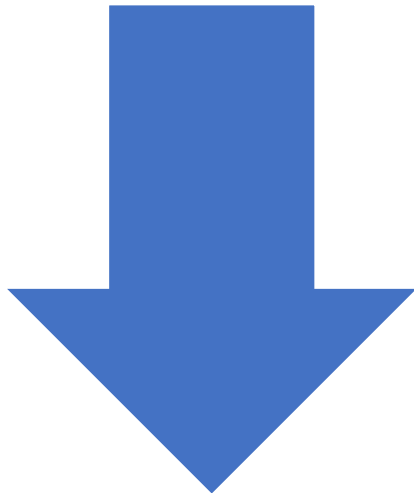
The use of cryptography raises concerns about whether individuals have a right to privacy in the first place. e.g. For example, some argue that the collection and storage of large amounts of personal data in the first place is a violation of privacy, regardless of whether that data is encrypted or not.

Ethical Aspects in Cryptography

The Right to Security



Cryptography can provide a high level of security for sensitive data by protecting it from unauthorized access and manipulation.



However, the use of cryptography can also **create a false sense of security**, leading individuals and organizations to take unnecessary risks or fail to properly secure their data in other ways.

Ethical Aspects in Cryptography (Contd.)

The right to access

- Cryptography can limit access to information, hindering research and other activities that require access to data.
- e.g. encrypted data may be inaccessible to researchers or journalists investigating a particular topic, limiting the ability to uncover important information.

The right to equality

- The use of cryptography can create a disparity between those who have access to encrypted data and those who do not.
- e.g. individuals or organizations with greater resources or technical expertise may be able to access encrypted data more easily than those without those resources.

Potential for abuse

- Cryptography can be used for illegal activities
- e.g. Money laundering or terrorism
- Creates challenging contexts where it is difficult to detect or prevent, such as in anonymous online transactions.

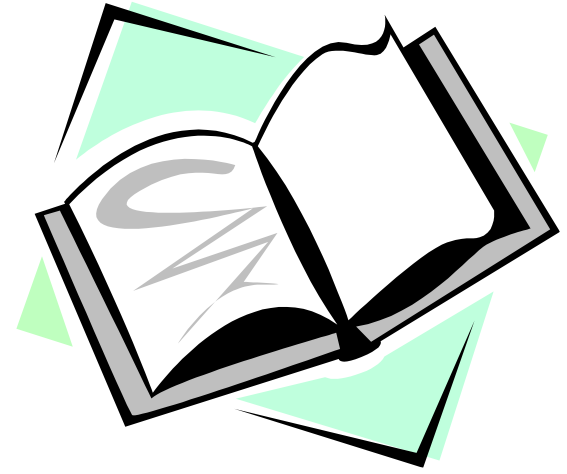
Examples of Ethical Cryptography Use Cases

- Applications :
 - Secure communication
 - Financial transactions
 - Healthcare data
 - Supply chain management
 - Voting systems
 - Digital Right Management (DRM)

Summary

- Confidentiality with symmetric encryption
 - Symmetric encryption
 - Symmetric block encryption algorithms
 - Stream ciphers
- Message authentication and hash functions
 - Authentication using symmetric encryption
 - Message authentication without message encryption
 - Secure hash functions
 - Other applications of hash functions
- Random and pseudorandom numbers
 - The use of random numbers
 - Random versus pseudorandom
- Public-key encryption
 - Structure
 - Applications for public-key cryptosystems
 - Requirements for public-key cryptography
 - Asymmetric encryption algorithms
- Digital signatures and key management
 - Digital signature
 - Public-key certificates
 - Symmetric key exchange using public-key encryption
 - Digital envelopes
- Practical Application: Encryption of Stored Data

Readings, Assignments and Labs



- **Readings**

- Computer Security - Principles and Practice, 4th Global Edition - William Stallings - Ch 2
- **Lab #02** - Lab Setup – Containerization + WebGoat installation
- Quiz $\Rightarrow$ next lecture (ISA)

Questions?

Thank You